



BKP: An R Package for Beta Kernel Process Modeling

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joint work with Kunhai Qing and Jin Xu

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Contents

1 Background

2 Summary



Background

- Estimating a **continuous probability function** from **binary** or **binomial observations** is fundamental in statistics and machine learning.
- Applications include:
 - Binary classification (MacKenzie et al., 2014; Wen et al., 2025)
 - Probability calibration (Sung et al., 2020; Dimitriadis et al., 2022)
 - Relative abundance modeling (Martin et al., 2020)
 - Longitudinal patient-reported outcomes (Najera-Zuloaga et al., 2018, 2019)
 - Infectious disease risk prediction (Diggle et al., 2007)
 - ...



Classical methods & Challenges

- **Parametric method:** logistic regression is **computationally efficient** but **limited in capturing complex nonlinear patterns**.
- **Nonparametric method:** Gaussian Process (GP) offers **flexibility and principled uncertainty quantification**, but **approximate inference** for binary (*non-Gaussian*) likelihood increases computational cost.



Beta Kernel Process

- **Beta Kernel Process (BKP)**: a scalable and interpretable nonparametric model for binomial probability surfaces (Goetschalckx et al., 2011; MacKenzie et al., 2014; Rolland et al., 2019)
- **Key features**:
 - **Computational efficiency**: closed-form posterior inference
 - **More transparency and straightforward**: beta-binomial conjugate pair
 - **Uncertainty quantification**: adaptive decision-marking
- **Dirichlet Kernel Process (DKP)**: a natural extension for categorical or multinomial data



Contents

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Summary

- Developed the **first publicly available BKP package** implementing BKP and DKP models
- **Methodological contributions:**
 - Provided a **Bayesian interpretation** of the BKP model
 - Proposed a **flexible framework for constructing priors**, including data-adaptive options
 - Developed a **robust and efficient hyperparameter optimization strategy** using LOOCV and multi-start reparameterization.



Future Development Directions

- **Model extensions:** Extend the BKP framework to multivariate, functional, and spatio-temporal data, as well as mixed-type covariates
- **Alternative likelihoods:** Explore likelihoods beyond the binomial family, e.g., negative binomial for over-dispersed counts, geometric for waiting-time modeling
- **Community contributions:** We invite developers to contribute via GitHub pull requests <https://github.com/Jiangyan-Zhao/BKP/pulls>



Resources

-  <https://cran.r-project.org/web/packages/BKP/index.html>
-  <https://github.com/Jiangyan-Zhao/BKP>
-  <https://arxiv.org/abs/2508.10447>





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